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A Novel Seismic and Sequence Stratigraphic Framework for the Progradational Lower Silurian Clastic Wedge in the Arabian Subsurface

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Following the well-known latest Ordovician glacial-terrestrial and glacial-marine phase, an early Silurian deglaciation brought mid-to-outer shelf facies, including a condensed Rhuddanian-age basal hot shale, throughout most of the North African and Arabian region (Jones & Stump, 1999; Konert et al., 2001; Sharland et al., 2001). The Lower Silurian hot shale, in particular, corresponds to organic-rich condensed marine facies with up to 8% total organic carbon and is the main source rock for the Paleozoic oil and gas accumulations in the region (e.g., Jones & Stump, 1999; Lüning et al., 2000; Konert et al., 2001).

After the earliest Silurian transgression, a very thick (several thousands of feet) coarsening-upward succession of Lower Silurian clastics prograded basinwards, forming shelf edge-scale clinoform-bearing regressive stratal architectures (Jones & Stump, 1999; Grabowski, 2005; Soliman & Balilah, 2015; Balilah, 2018). As a consequence, fluvio-deltaic, shallow- to open-marine ('shelfal' and 'shelf-margin') sandstone-rich settings developed towards paleo-landward directions. At the same time, deeper-marine ('delta-toe' or offshore) clay-prone conditions were established in two rapidly subsiding distal sagging depocenters, developed following the Taconic Uplift (e.g., Jones & Stump, 1999; Konert et al., 2001; Soliman & Balilah, 2015). One of these broad tectonic basins, known as 'Qalibah Trough', formed in central and southern Arabia from the Late Ordovician; the latter, known as 'Taymah Trough', formed in north-western Saudi Arabia from the Early Silurian (Jones & Stump, 1999).

In and around the Qalibah Trough, the Lower Silurian succession (LSS) is an offshore mudstone-rich siliciclastic wedge, showing an overall coarsening-upward log motif and progradational clinoform-bearing seismic architecture (Soliman & Balilah, 2015), representing a part of the AP3 Megasequence of Sharland et al. (2001) (Table 1). Several sandstone-rich members often occur in the middle and upper part of the LSS, corresponding to Rhuddanian and Aeronian ages, respectively (e.g., Miller & Melvin, 2005; Soliman & Balilah, 2015), but their presence and distribution has not been completely clarified. Lower Silurian offshore turbiditic-influenced and lower-shoreface siltstone, silty sandstone, and fine-grained sandstone facies were described from the Tabuk outcrops (Abdulmutalib et al., 2022). Sigmoidal shelf-edge scale clinoforms with ascending trajectories, as well as steeper shelf-edge scale clinoforms with low-angle or descending trajectories were identified by Soliman & Balilah (2015) and Balilah (2018). These authors also speculated on the presence of shelf-edge and slope channels and possible deep-basin submarine fans.

INTRODUCTION

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Table 1—Lower-Middle Silurian Biostratigraphic/chronostratigraphic zones and surfaces utilised in this work. The stratigraphy is modified after Paris et al. (1995), Miller & Melvin (2005) and Cesari et al. (2018). FDO means 'first downhole occurrence' of a given biostratigraphic taxon.

MEGA-SEQUENCES	CHRONO STRATIG RAPHIC UNITS	SAUDI ARAMCO PALYNO ZONES	CHITINO-ZOAN ZONES	CHRONOSTRATIGRAPHIC SURFACES			
				No	Main Palynostratigraphic markers (FDOs)	Secondary Palynostratigraphic markers (FDOs)	Age of the surface
LDepositional Sequence DS1 (sensu Sharland et al., 2001) [part of Megasequence AP3 sensu Sharland et al., 2001]	LOWER TELYCHIAN	S2A-I	<i>T. obtusa</i> Zone p.p.	Early Acadian SB30 Sequence Boundary (major order)			
				01	<i>Plectochitina longicornis</i>	-	Middle Telychian
			<i>A. macclurei</i> Zone	02	<i>Angochitina macclurei</i>	<i>Fractoricoronula trispina</i>	Early Telychian
	UPPER AERONIAN	S2A-II	<i>A. hemeri</i> Zone	03	<i>Angochitina hemeri</i>	-	Top Aeronian
				Late Aeronian Transgressive Surface			
				04	<i>Rugosphaera agglomerata</i> (sl. younger) <i>Belonechitina arabiensis</i> (sl. older)	<i>Tylotopalla caelamenicutis</i> (intermediate age?)	Late Aeronian
				Mid Aeronian SB20 Sequence Boundary			

MEGA-SEQUENCES	CHRONOSTRATIGRAPHIC UNITS	SAUDI ARAMCO PALYNO ZONES	CHITINOZOAN ZONES	CHRONOSTRATIGRAPHIC SURFACES			
				No	Main Palynostratigraphic markers (FDOs)	Secondary Palynostratigraphic markers (FDOs)	Age of the surface
	LOWER AERONIAN	S2B	<i>S. solitudina</i> Zone	05	<i>Sphaerochitina solitudina</i>	-	Middle Aeronian
			<i>C. alargada</i> Zone	06	<i>Conochitina alargada</i>	<i>Angochitina qusaibaensis</i> (slightly younger)	Early Aeronian
				Maximum Flooding Surface MFS S10 of Sharland et al. (2001)			
	RHUDDANIAN	S2C	<i>A. udayanensis</i> Zone	07	<i>Ancyrochitina udayanensis</i>	<i>Oppilatala conisphaera</i> <i>Eupoikilofusa praecurvata</i>	Top Rhuddanian
				Top Rhuddanian Transgressive Surface			
				Late Rhuddanian Sequence Boundary			
				Middle Rhuddanian Transgressive Surface			
			<i>L. nuayyimensis</i> Zone	08	<i>Lagenochitina nuayyimensis</i>	-	Middle Rhuddanian
			<i>B. postrobusta</i> (<i>pseudarabiensis</i>) Zone	09	<i>Belonechitina pseudarabiensis</i>	<i>Belonechitina postrobusta</i>	Early Rhuddanian
				Top Early Rhuddanian Sequence Boundary			
			<i>S. fragilis</i> Zone	10	<i>Spinachitina fragilis</i>	-	Earliest Rhuddanian
	HIRNANTIAN	O1A	Base Silurian Transgressive Surface				
			Base Hirnantian SB10 Sequence Boundary (major order)				

Here, we integrate well and seismic data to attempt to describe and predict the extent of these sandstone-rich sub-units in the Arabian sub-surface.

METHODS

Palynologically-constrained sequence stratigraphic regional well correlation panels were integrated with 2D and 3D seismic stratigraphic observations. Numerous wellbores and thousands of kilometres of seismic lines have been interpreted.

Palynological data from cores and cuttings were categorized into high-confidence and medium-confidence datasets based on sample quality and origin. In particular, the biostratigraphic framework already discussed by Miller & Melvin (2005) has been further refined here into ten Rhuddanian, Aeronian and Early-Middle Telychian biostratigraphically-tied chronostratigraphic surfaces (Table 1).

The raw palynological ‘first downhole occurrence’ (FDO) dataset has been refined to avoid spurious events through the following methodology. Firstly, all out-of-sequence palynological events were deleted from the dataset.

Secondly, all shallowest co-occurring multiple palynological events have also been deleted. These are deemed to be no real FDOs/unconformities, but simply representing the shallowest sample analysed in a given well, where all taxa are necessarily identified together for the first time.

Mature, organic-rich, marine shales are defined petrophysically based on low sonic and bulk density values and very high gamma ray, uranium, neutron and resistivity values. This is due to the low-density,

non-conductive, high hydrogen index and high authigenic uranium content of most mature marine organic matter (Lüning et al., 2000; Rider & Kennedy, 2011).

Synthetic seismic and VSPs were utilized to identify real seismic reflectors and distinguish them from multiples and artefacts. The scale of observations ranged from regional (100s km) to more local (10s km). The classical sequence stratigraphic methodologies outlined by Mitchum et al. (1977), Van Wagoner et al. (1988, 1990), Neal & Abreu (2009) and Abreu et al. (2010) were applied to this study. In particular, a maximum regression (=transgressive) surface is equated in seismic to a paleo- seaward to paleo-landward switchover in clinoforms trajectories (Fig. 1). The opposite switchover is equated to a maximum retrogradation (=maximum flooding) surface, and is often further highlighted by reflector downlap terminations (Fig. 1). Finally, an abrupt downward/basinward shift in clinoforms breaks marks the sequence boundary, and is further highlighted by basinal onlap terminations and proximal erosional truncation of reflectors (Fig. 1). The lowstand system tract, underlain by a sequence boundary and overlain by a transgressive surface, is the unit for which the maximum amount of porous sandstone is expected to be delivered to the deep basin, beyond the shelf-edge break (Fig. 1).

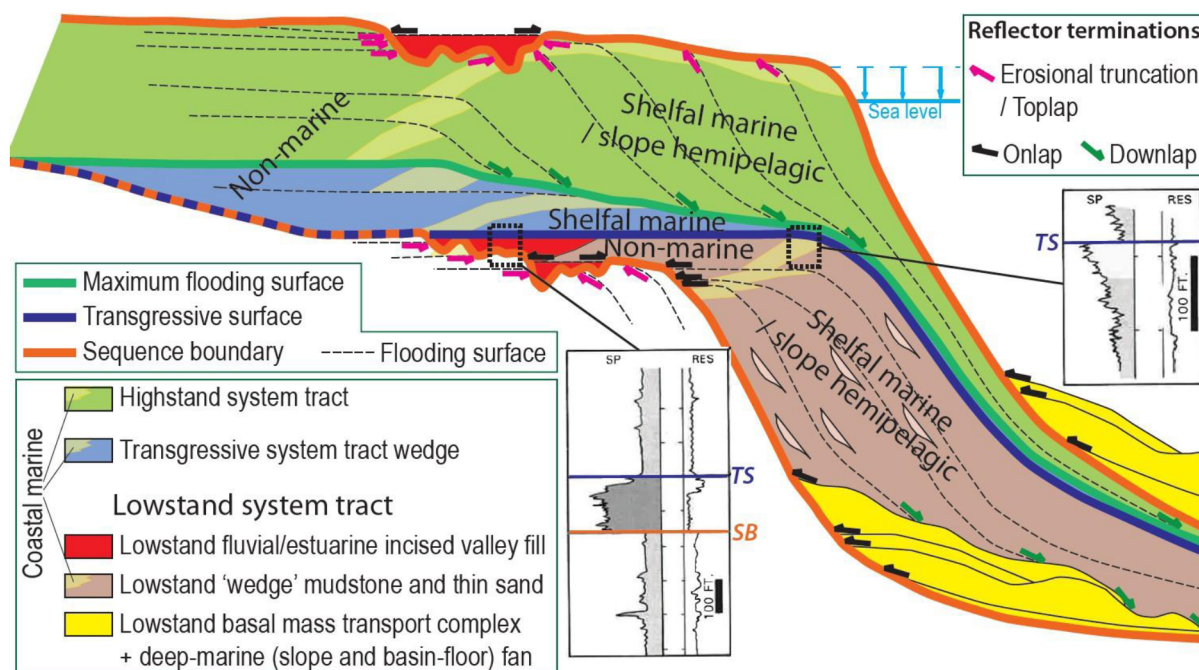


Figure 1—Sketch depicting a classical 'Exxonian' depositional sequence, bounded by two subsequent sequence boundary surfaces at its base and top. Inside the depositional sequence, the maximum flooding surface and the transgressive surface are also defined. The lowstand system tract is the unit for which a maximum delivery of porous sandstones is expected to the deep-water slope and basin-floor. Sketch modified after: Van Wagoner et al. (1988, 1990), Nel & Abreu (2009), Abreu et al. (2010).

Wherever the seismic imaging allows, sequences show multiple clinoform surfaces. These are ubiquitous inclined stratal, chronostratigraphic and depositional surfaces (e.g., all the flooding surfaces, sequence boundaries, transgressive surfaces and maximum flooding surfaces in Figure 1), corresponding to "frozen" palaeo-bathymetric and palaeo-topographic profiles (Patrino & Helland-Hansen, 2018). Shelf-edge scale clinoforms, in particular, are defined as clinoforms showing an overall relief of hundreds of metres (c.f., Patrino & Helland-Hansen, 2018). These clinoform types are typically composed of muddy, offshore-to-offshore-transition lithotypes, such as those composing the bulk of the Arabian LSS (Patrino & Helland-Hansen, 2018). The only exception is represented by relatively thinner intercalations of sandier units that developed whenever the deltaic sediment input-point managed to prograde close to the shelf-edge, forming a hybrid 'shelf-edge delta' (Patrino & Helland-Hansen, 2018). The study of shelf-edge scale clinoforms

from all over the world revealed that changes in shelf-edge clinoform trajectories and geometries may be significant features when it comes to predicting the development of deeper-marine porous and permeable units (e.g., [Helland-Hansen & Hampson, 2009](#); [Patrino & Helland-Hansen, 2018](#)).

RESULTS

The Rhuddanian part of the LSS

The Rhuddanian part of the LSS is partly organic-rich, shows a predominantly transgressive log motif and seems to infill underlying faulted and folded palaeo-morphologies ([Fig. 2](#)).

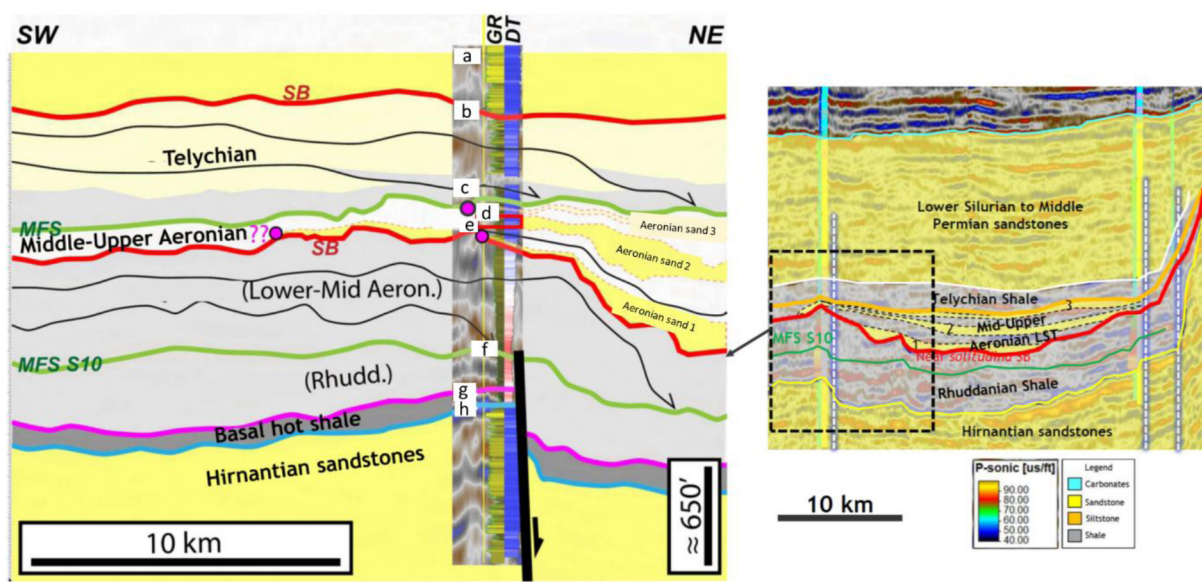


Figure 2—Seismic cross-section and geoseismic section crossing a well from our dataset, depicting the key seismic stratigraphic features discussed in this article, particularly for the Aeronian part of the LSS wedge. Tops: (a) Lower Devonian TS, (b) Upper Silurian SB, (c) Intra-Telychian, (d) Late Aeronian, (e) Mid Aeronian SB, (f) MFS S10, (g) Lower Rhuddanian hot-warm shale, (h) Base Silurian TS.

The lower Rhuddanian, in particular (corresponding to the *B. postrobusta* biozone – [Table 1](#)), often shows a basal ‘hot shale’ and ‘warm shale’ wireline log response, at least in its lowermost part ([Fig. 4](#) – see ‘Methods’ section). The thickness of the hot shale seems to be somehow related to the inferred Early Rhuddanian sedimentation rate, with maximum thickness reached around ‘optimum’ sedimentation rate values of 25-35 m/Myr ([Figs. 3-4](#)).

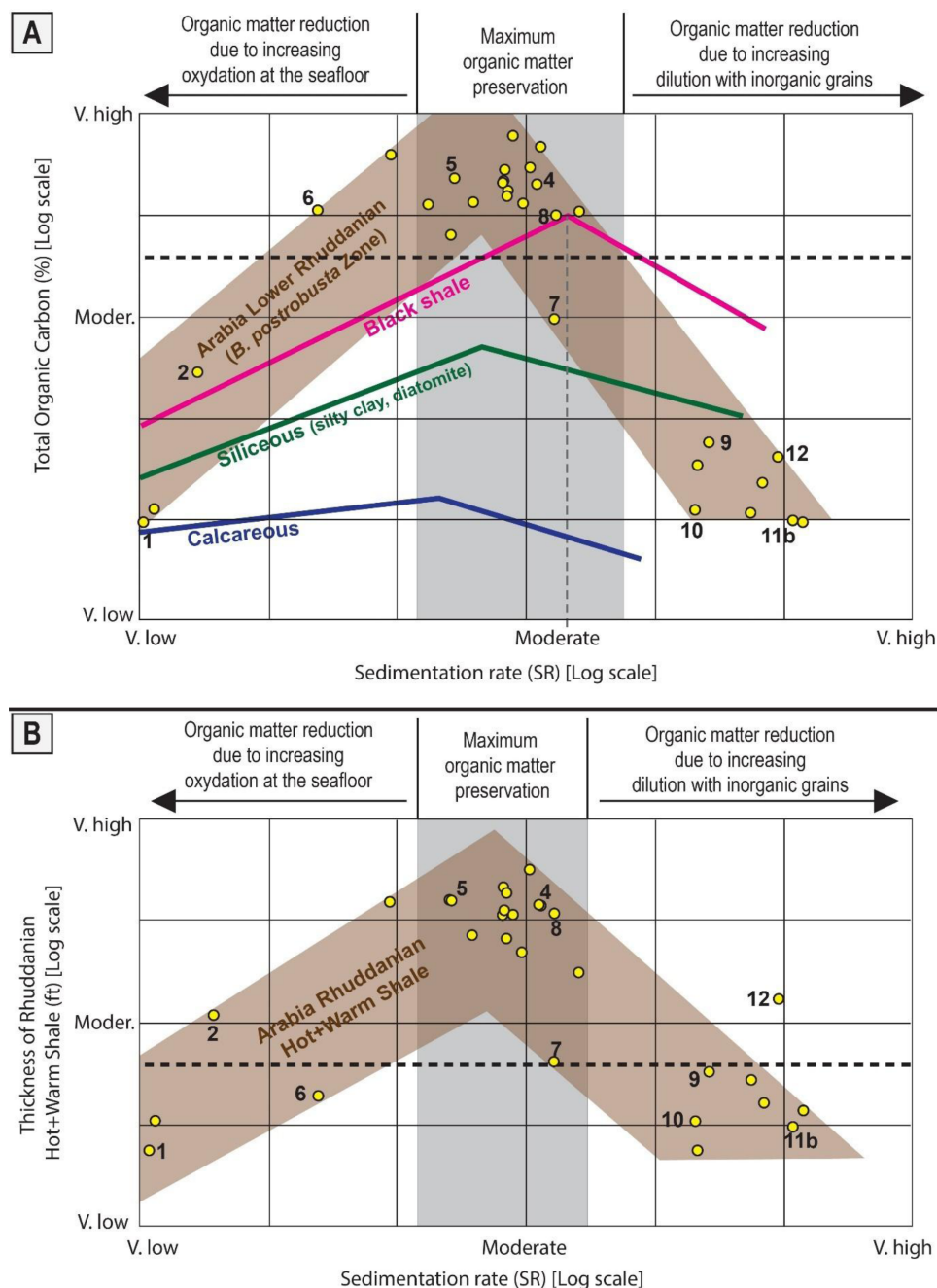


Figure 3—(A) Log-log trends of Sedimentation rate versus Total Organic Carbon (TOC) for black shale, siliceous and calcareous units in global Jurassic to Recent Deep Sea Drilling Project cores (modified after [Ibach, 1982](#)). The maximum organic matter preservation is reached for an optimum/intermediate range in sedimentation rates. On top of this same graph, we have cross-plotted yellow dots. Each indicates a single well in our dataset, with its value of average sedimentation rate for the Early Rhuddanian *B. postrobusta* (*B. pseudarabiensis*) palynostratigraphic Zone in m/Myr (x axis, with resolution dependent on the palynological sampling rate) versus its average petrophysically-inferred TOC for the same biozone. If the Lower Rhuddanian hot+warm shale unit has been deposited and preserved, it will be at the base of this biozone (also see [Table 1](#) and [Figure 4](#)). The numbers refer to the wells shown in [Figure 4](#). **(B)** Cross-plot of the average sedimentation rate for the Early Rhuddanian *B. postrobusta* (*B. pseudarabiensis*) palynostratigraphic Zone in m/Myr (x axis, with resolution dependent on the palynological sampling rate) versus total thickness of the petrophysically-defined Lower Rhuddanian Hot Shale + Warm Shale interval in feet (y axis). See [Table 1](#) for the palynostratigraphic definition of the Zone. The numbers refer to the wells shown in [Figure 4](#). N.B. The points shown here represent only wells for which the FDO of *B. postrobusta* (or *B. pseudarabiensis*) are present not out-of-sequence with the other palynological events, and are not the shallowest identified palynological event, as these are probably not real FDOs. We deem the thickness of the hot+warm shale (part B) to be a good proxy for TOC (part A), because these intervals were defined on the basis of a minimum petrophysically-calculated total organic carbon cut-off of 2.0% and above (see 'Methods').

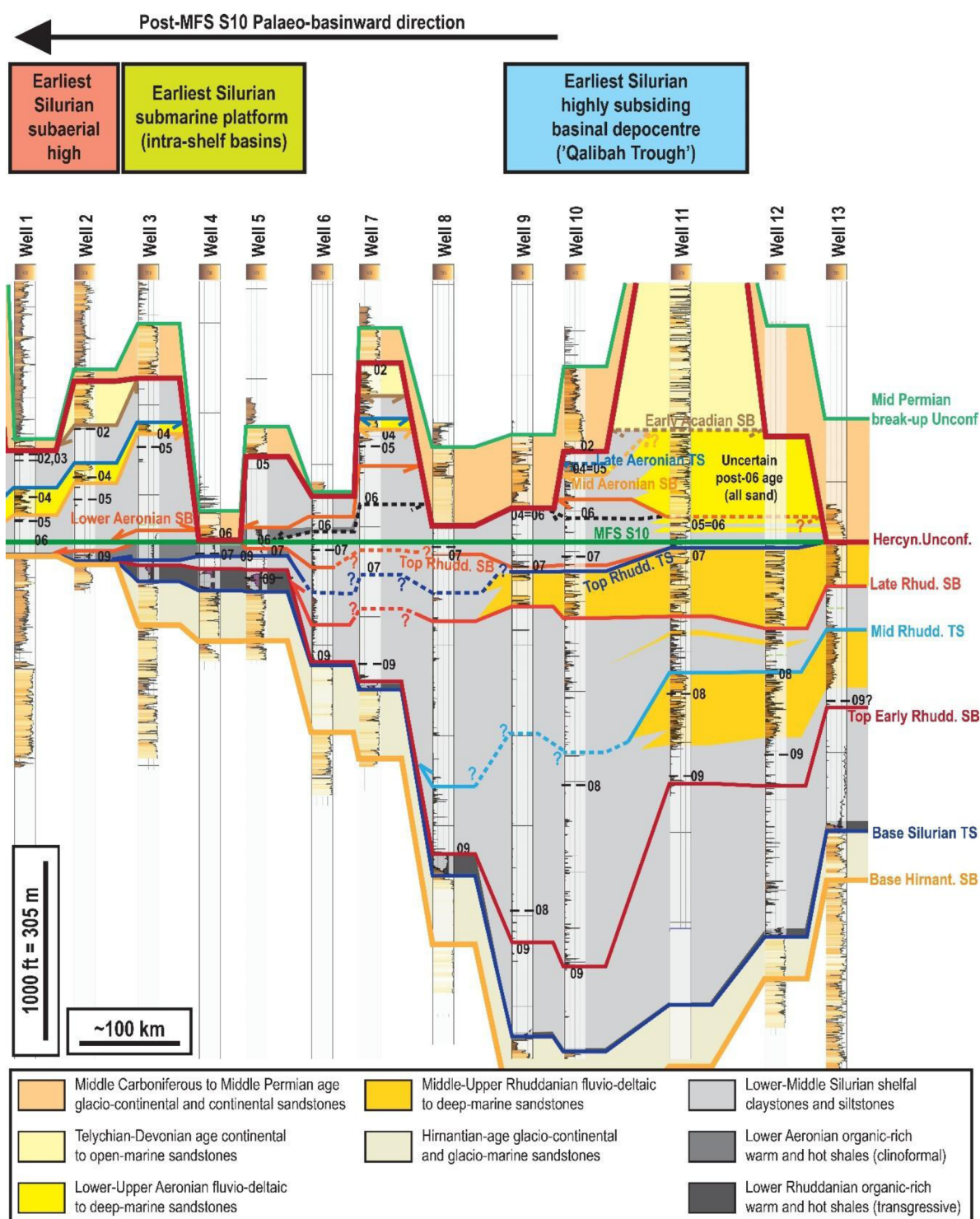


Figure 4—Regional well-correlation panel illustrating the key features discussed in this paper, including the variability in the thickness and number of hot-warm shale intervals in the different tectono-stratigraphic settings and the stratigraphic architecture of the intra-Rhuddanian and intra-Aeronian sandstone units. The logs shown are petrophysically-derived 'volume sand' (yellow to grey color scale) and total organic carbon above 2.5% (black-purple color scale). The numbers refer to the stratigraphic depths of the palynostratigraphic events of Table 1. Also see Table 1 and text for definition of the stratigraphic surfaces shown. The interval between the Early Rhuddanian SB and the Base Silurian TS is largely equivalent to the B. postrobusta (pseudarabiensis) Zone (Table 1). Therefore, its thickness has been utilised to calculate the Early Rhuddanian sedimentation rate shown in the x axis of the cross-plot of Figure 3. Standard colors have been used for the surfaces shown in this figure: red-orange for sequence boundaries, green for maximum flooding surfaces and blue for transgressive surfaces (also see Fig. 1). The well numbers from this figure have also been shown in the cross-plots of Figure 3.

The overall transgressive Rhuddanian trends are seemingly interrupted in the Middle Rhuddanian *L. nuayyimensis* Zone (Table 1). The thickness of this zone seems to vary widely, from completely absent on the basin flanks to very thick in the basinal depocenters. This is also shown by the extreme thickness variations of the sedimentary unit between the Late Rhuddanian and the Top Early Rhuddanian sequence boundaries in Figure 4. At this very time, during the Middle-Upper Rhuddanian *L. nuayyimensis* and *A. udayanensis* zones, intra-Rhuddanian sandstone-rich units were also deposited in the basinal depocenters (Fig. 4).

The Aeronian-Lower Telychian part of the LSS

A main transgressive-to-regressive switchover in log motifs occurs around the earliest Aeronian, typically just above the FDO of *A. udaynanensis* (Table 1; Figs. 2, 4). This surface, roughly time-equivalent with the S10 maximum flooding surface of Sharland et al. (2001), corresponds in seismic to a surface on top of which shelf-edge scale Silurian clinoforms are seemingly downlapping (Fig. 2 – also see the down-dip termination of Clinoform 06 in the well correlation panel of Figure 4). These clinoforms have been mapped from both our geological and geophysical datasets and show an overall northeastward basinward-progradation direction; however, around the eastern margin of the basin, a few smaller westward-prograding clinoforms were mapped, suggesting minor counter-regional sediment input points.

Just above the FDO of *S. solitudina*, in the Middle Aeronian and Lower Telychian part of the LSS, the gradually cleaning-upward regressive log motifs of the LSS are interrupted by a Middle Aeronian candidate sequence boundary unconformity, in the lowermost part of the *A. hemeri* palynostratigraphic zone. In particular, in wells drilled in the south-western sides of the basin (i.e., towards a paleo-landward direction), a Middle Aeronian unconformity is marked by an interpreted mudstone-on-mudstone erosional surface with missing palynostratigraphic markers (in particular, surfaces 4 and 5 of Table 1; also see Well 10 in Figure 4 and Miller & Melvin, 2005), suggesting a possible lowering in relative sea-level and erosional truncation terminations developed around the shelfal parts of sequence boundaries (e.g., Fig. 1). Further paleo-seaward (i.e., to the north and north-east), this shelfal unconformity turns into a correlative conformity just over the FDO of *S. solitudina* (Figs. 2, 4; Miller & Melvin, 2005). Above this basinal sequence boundary, the gradual LSS cleaning-upward regressive log motifs breaks down, and two or three sharp-based Middle Aeronian-Lower Telychian deeper-marine sandstones were deposited, mainly in the *A. hemeri* palynostratigraphic Zone, between surfaces 3 and 5 of Table 1 (Fig. 4; also see Miller & Melvin, 2005). In cores, these sharp-based sandstones can be interpreted as inter-stratified storm-generated bottom currents and turbidites that moved down the Early Silurian depositional slope, transitioning to more proximal shelfal and marginal-marine sandstones along a paleo-landward direction (e.g., Soliman & Balilah, 2015).

In seismic, the Middle Aeronian surface at the base of the Middle-Upper Aeronian wedge is marked by the possible onlap terminations of several younger reflectors (Fig. 2), thus representing a candidate sequence boundary (c.f., Van Wagoner et al., 1988, 1990; Figs. 1-2, 4). Here, we highlight a discrete region where the best Middle Aeronian-Lower Telychian sandstones were deposited, just above the Middle Aeronian (*S. solitudina*) clinoform sequence boundary. The south-westward edge of the Middle-Upper Aeronian wedge, marked by the uppermost onlap reflector terminations in seismic (Fig. 2), may represent the landward-most extent of most Middle Aeronian-Lower Telychian turbiditic sandstones. Basinward of this point, all sandstone members were encountered by most wells, showing minor thickness variations over a large area, with their petrophysical properties suggesting a possible lateral transition to more distal facies (Fig. 2, also see the area to the left-hand side of Well 10 in Figure 4). Such stratigraphic architecture suggests a newly described candidate lowstand turbiditic wedge within the LSS.

The Middle-Upper Aeronian seismic wedge is topped by a new transgressive-regressive log trend that in seismic corresponds to the progradation of a Middle Silurian clinoform-bearing clastic wedge (Fig. 2). Towards the basin edges, the Silurian succession is often truncated by the Pre-Late Silurian ('Early Acadian') Unconformity, which is in turn overlain by continental clastics and again eroded by the Hercynian Unconformity, further complicating the correct interpretation of the Silurian geological evolution.

DISCUSSIONS

The Lower Rhuddanian organic-rich shale interval

Our analysis of the thickness distribution of hot and warm shale intervals associated with the base of the LSS (Lower Rhuddanian) shows that maximum organic-rich shale thicknesses are reached around ‘optimum’ sedimentation rate values of 25–35 m/Myr (Figs. 3–4). This is largely in line with global trends between organic matter preservation and sedimentation rates (Ibach, 1982; Fig. 3). For higher sedimentation rates, the hot shale thickness decreases significantly, probably due to organic matter sedimentary dilution in the basinal depocenters; for slower rates, oxidation of organic matter would increase, causing minor organic preservation (Ibach, 1982).

It is possible that the wells characterized by very low sedimentation rates in the Early Rhuddanian correlate with the latest Ordovician-Rhuddanian paleo-physiographic highs, on top of which the basal hot and warm shales thin by onlap (Figs. 2, 4; also see Grabowski, 2005). According to the pan- northern Gondwana model of Lüning et al. (2000), the deposition of the Rhuddanian hot shales is discontinuous, and their extent and thickness are controlled by the Early Silurian paleo-relief, in turn shaped by Late Ordovician glacial processes superimposed on larger structural undulations, likely inherited by pre-Silurian compressional and extensional tectonic phases. Accordingly, on shelves, the deposition of the hot shales during the early transgression phase took place only in paleo- depressions, whereas major paleo-highs (likely still above sea-level or nearly above it) were only reached later by the Early Silurian transgression, and therefore completely lack hot shale deposition, with the earliest deposited Silurian sediments in these areas being later lean shales (Lüning et al., 2000; Fig. 4).

On the other side, Jones & Stump (1999) recognized that the most anoxic organic-rich Rhuddanian hot shale was deposited away from the two regional highly-subsiding basinal depocenters of the Qalibah and Taymah troughs (e.g., thick hot shale to the east of the Qalibah Trough), where organic matter produced in the water column was diluted by the high sedimentation rates (Figs. 3–4). Jones & Stump (1999), moreover, suggested that the organic-rich shales in wells situated in the axes of the Qalibah and Taymah troughs show sharp/erosional tops, whereas in wells away from these regional depocenters, the organic-rich shales progressively grade upward into leaner shale. The sharp/erosional hot shale tops in the depocenters have been interpreted to reflect organic-matter reworking and oxidization by the action of turbidites, contourites and storm-generated bottom currents (c.f., Jones & Stump, 1999), further reducing the overall thickness or the lowermost Silurian organic-rich hot and warm shale in these very highly-subsiding areas.

In conclusion, we can propose that the regions with very low (or no) Early Rhuddanian sedimentation rate and negligible hot shale deposition due to high organic matter oxidation (e.g., dots to the left hand-side of the cross-plot of Figure 3) correspond to nearly-subaerial paleo-highs, which is consistent with the shelfal model of Lüning et al. (2000) (Fig. 4). On the other hand, the areas with very fast Early Rhuddanian sedimentation rate and thin basal hot shale due to elevated organic matter dilution (e.g., dots to the right hand-side of the cross-plot of Figure 3) correspond to basinal depocenters beyond the shelves, which is consistent with the observations of Jones & Stump, 1999 (Fig. 4). Finally, optimum intermediate Early Rhuddanian sedimentation rates (c. 25–35 m/Myr – c.f., dots in the center of the cross-plot of Figure 3) were likely reached in shelfal paleo- depressions, such as structural intra-shelf basins or inherited Ordovician glacial outwash valleys (Fig. 4), which is consistent with the models of both Jones & Stump (1999) and Lüning et al. (2000).

The Middle-Upper Rhuddanian sandstones

The Rhuddanian unit comprises the bulk of the LSS thickness, particularly in the highly-subsiding regional depocenters (Fig. 4). Possible Rhuddanian phases of basin growth and basin flank uplifts are shown by the large variations in thicknesses of the intra-Rhuddanian biozones, interrupting the overall transgressive Early Silurian trend (Fig. 4). This is particularly true when it comes to the Middle Rhuddanian *L. nuayyimensis*

Zone, which is completely missing on the shelfal highs beyond the Qalibah Trough (*sensu* Jones & Stump, 1999), whereas becomes very thick in the basinal depocenters of the Qalibah Trough, where time-equivalent middle-upper Rhuddanian basinal sandstone units were also deposited (Soliman & Balilah, 2015; see marker 08 in Figure 4 and Table 1). These extreme regional variations in thickness for the Middle Rhuddanian units can be interpreted as a rejuvenation of the basin physiography during this time (i.e., relative uplift of basin flanks and enhanced subsidence of the Qalibah Trough), causing an overall fall in relative sea-level and onset of lowstand deposition, including lowstand sandstones in the basins (Fig. 4).

In the same depocentral area, the subsequent Middle-Upper Aeronian sandstone shows more proximal (fluvio-deltaic and ‘slope-deltaic’) shelf-margin facies (Soliman & Balilah, 2015), whereas the time-equivalent Middle-Upper Aeronian sharp-based deeper marine sandstone facies described by Miller & Melvin (2005) and in sections 3.2 and 4.3 of this article, were typically deposited more paleo-seawards, to the north and north-east of the Qalibah Trough depocenter, reflecting the long-term progradational migration of the LSS depositional systems and the largely Rhuddanian-age infilling and levelling off of the Qalibah Trough depocenter (Fig. 4).

The stratal architecture of the interpreted intra-Rhuddanian markers above the basal hot shale suggests a gradual basinwards progradation of the depositional systems (Fig. 4). The markers form broad shelf-edge scale clinoform sets, which prograded through the entire width of the Qalibah Trough beginning from its paleo-landward margin. They eventually encroached onto the opposite basin flank, where they showed signs of upslope/landward progradation, with hybrid onlap/downlap stratal terminations (Fig. 4). This ‘upslope climbing clinoform’ style is typical of some sediment- overfilled epicontinental basins: a very similar analogue is represented by the lower Quaternary Eridanos shelf-edge delta of the North Sea, (c.f., Patruno et al., 2020).

The earliest Aeronian maximum flooding surface and the Early Aeronian highstand

After the rate of relative sea-level rise begins to decline, the main transgressive-to-regressive switchover in log motifs occurred around the earliest Aeronian, typically just above the FDO of *A. udaynanensis* (Table 1) (Figs. 2, 4), as already observed by Miller & Melvin (2005). This surface is largely time-equivalent with the S10 maximum flooding surface postulated by Sharland et al. (2001) and corresponds in seismic to a surface on top of which shelf-edge scale Silurian clinoforms are seemingly downlapping (Fig. 2; see the distal termination of the Clinoform 06 in Figure 4). This conforms to the seismic stratigraphic definition of ‘maximum flooding surface’ (e.g., Van Wagoner et al., 1988, 1990; Fig. 1). Highstand mud-prone cyclothems that downlap onto the MFS S10 surface were previously inferred based on palynostratigraphic data alone (Miller & Melvin, 2005). These clinoforms comprise basinward-thinning progradational shale-siltstone units (c.f., Grabowski, 2005) and show an overall northeastward basinward-progradation direction (Fig. 2), which is consistent with the analyses of Miller & Melvin (2005) and Soliman & Balilah (2015). At the same time, sandstone-rich highstand delta systems started to prograde from the northern Gondwana coasts into the shaly Silurian shelfal basins (Lüning et al., 2000; Fig. 4).

In the northern Arabia basin, which is sited in a more paleo-seaward location than our study region, the maximum extent of a younger hot-shale interval has been dated to the Middle Aeronian, which is also consistent with the age of the S10 maximum flooding event (Hayton et al., 2017). Indeed, apart from the basal (lower Rhuddanian) transgressive hot shale that laps onto highs and progressively becomes very condensed/thin distally, hot shales with clinoformal geometries have been identified higher up in the Lower-Middle Silurian stratigraphy across some of the more distal basins of Northern Gondwana (Grabowski, 2005; Fig. 4). These higher ‘clinoformal’ hot shales are likely related to maximum flooding surface clinoforms, such as the S10 surface (Fig. 4).

The Middle-Upper Aeronian sandstones

The development of the Middle-Upper Aeronian erosionally-based turbiditic and tempestitic sandstones that initially formed after the FDO of *S. solitudina* and then immediately after the FDO of *R. agglomerata* (c.f., surfaces 04 and 05 in Table 1 and Figures 2 and 4), was already noted in Central Arabia, albeit at a more local scale than in our work, by the palynostratigraphic and sedimentological analyses of Miller & Melvin (2005). These authors related the onset of the deposition of the lowstand Middle-Upper Aeronian sandstones to a middle Aeronian global sea-level fall. This event has also been identified in northern Arabia, which is situated in a more paleo-seaward setting than our study region, with a largely time-equivalent peak in shale sedimentation rate and abundant interbeds of siltstones and very-fine to fine-grained sandstones in coarsening-upward parasequences documented by Hayton et al. (2017).

Following the sedimentary flattening of the inherited Ordovician-earliest Silurian paleophysiographic undulations by the deposition of the very thick Rhuddanian clastic wedge, the Aeronian units were deposited with a clear MFS S10-downlapping clinoformal architecture (Figs. 2, 4). Progressively younger clinoforms prograded paleo-seawards, until reaching former earliest Silurian subaerial highs lacking any hot shale deposition, and that had now become rather deep- water Aeronian marine basins (e.g., Well 1 and Well 2 in Figure 4). Renewed thickening of Aeronian- Telychian units in the Qalibah Trough (e.g., Wells 11 and 12 in Figure 4) is suggestive of a new phase of enhanced subsidence and basin growth occurring in this regional depocenter following a relative tectonic quiescence and passive sedimentary infilling during the Late Rhuddanian.

The Pre-Late Silurian Unconformity (or ‘Early Acadian Unconformity’) marks the top of the LSS and is overlain with lowstand sandstones (e.g., Lüning et al., 2000; Fig. 4). This sequence boundary is equivalent to the regional Acadian phase of deformation (Jones & Stump, 1999; Table 1; Fig. 2), which is an earlier deformation compared to the Carboniferous-age event that gave rise to the widespread Hercynian Unconformity (c.f., Sharland et al., 2001; Konert et al., 2001).

CONCLUSIONS

The deposition of the largely offshore claystone and siltstone-prone Lower Silurian clastic wedge (LSS) in the subsurface of central Arabia has been summarised below.

1. The end-Rhuddanian de-glaciation causing earliest Silurian transgression and deposition of a transgressive basal Silurian hot shale unit showing significant variations in thickness depending on the tectono-stratigraphic setting of each area. In nearly-emergent paleo-highs, the lower Rhuddanian was not deposited at all or is very thin, whereas in highly-subsiding basinal depocenters (e.g., Qalibah Trough) the basal hot shale is very thin because it is highly diluted by high inorganic sedimentation rate. Conversely, very thick hot shales were deposited in intermediate submarine platforms (‘intra-platform basins’), characterized by optimum sedimentation rates for enhanced organic matter preservation.
2. A middle Rhuddanian rejuvenation in basinal physiography (marginal uplifts and depocentral subsidence) caused a fall in relative sea-level and the onset of lowstand settings. The Middle-Upper Rhuddanian units are therefore only present in the Qalibah Trough depocenter, with enhanced delivery of sandstones (‘Rhuddanian sands’) to the basin floors. This lowstand phase interrupted the regional Rhuddanian transgressive trend.
3. The very thick Rhuddanian wedge comprises the majority of the LSS unit thickness, completely infilling the very thick Qalibah Trough depocenters. Following the deposition of an earliest Aeronian maximum flooding surface (MFS S10), the regional physiography looked much more uniform than at the beginning of the Silurian times.

4. Highstand conditions were established in the Early Aeronian, with shelf-edge scale claystone-prone clinoforms progressively prograding basinwards and downlapping onto the underlying MFS S10 surface.
5. In the Middle Aeronian, a eustatic sea-level fall ensued, causing the development of lowstand sedimentation once more. In the paleo-basinward part of the study region, several sharp-based interbedded turbiditic and tempestitic Middle-Upper Aeronian sandstone units were deposited above a basal sequence boundary, just after the FDO of *S. solitudina*. Moving towards a paleo-landward direction, these units progressively amalgamate together and onlap against the middle Aeronian paleo-slope. However, further paleo-landward, more proximal time-equivalent fulvio-deltaic and shallow-marine sandstone facies were deposited, for example in the depocenter of the Qalibah Trough. The middle-upper Aeronian lowstand clastic wedge is topped by a Late Aeronian transgressive surface and by Telychian shales.
6. In the Telychian, a new sequence boundary brought an end to the deposition of the widespread claystone- and siltstone-rich LSS unit. This unconformity is equivalent to the Early Acadian regional deformation phase. The Early Acadian sequence boundary is in turn overlain by continental and shallow-marine sandstones.
7. Parts of the LSS wedge and overlying upper Silurian-Devonian succession was then eroded away by the regional major Hercynian Unconformity in the Middle Carboniferous to Middle Permian time, particularly towards the basin margins, further complicating the interpretation of the Silurian geological evolution.

The south-westward edge of the Middle-Upper Aeronian wedge, marked by the uppermost onlap reflector terminations in seismic, may represent the landward-most extent of most Middle-Upper Aeronian turbiditic sandstones. Basinward of this point, all sandstone members were encountered by most wells, showing minor thickness variations over a large area, with their petrophysical properties suggesting a possible lateral transition to more distal facies. Such stratigraphic architecture is suggestive of a newly-described candidate lowstand turbiditic wedge within the lower-middle Silurian clastic wedge.

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